Impact of Women's Reservation on Public Sector Employment: Evidence from Punjab

Term Paper

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Instructor: Prof. Sevrin Waights

Name: Arunima Marwaha

Matriculation Number: 644552

Study Program: M.Sc. Economics

Email Address: arunima.marwaha@student.hu-berlin.de

Humboldt-Universität zu Berlin

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1 Introduction

1.1 Context & Policy Background

In India, both central and state governments implement reservation policies in public institutions to uplift historically disadvantaged groups. These reservations apply across a range of institutions, including parliament, local bodies, educational institutions, and civil services and are typically targeted towards minority groups defined by gender, caste, disability status. However, their effectiveness remains an empirical question.

In October 2020, the Government of Punjab (a state in northern India) introduced a policy of 33% reservation for women in direct recruitment to Punjab Civil Services. The quota applies to all establishments in the public sector[1] in Group 'A', Group 'B', Group 'C', and Group 'D' services.[2] The reservation policy follows the *"horizontal and compartmentalised"* rule, which means that it applies within all existing social categories (for example, marginalised castes and people with disabilities) instead of replacing them.

1.2 Research Question

This paper studies the causal impact of the 33% reservation quota for women in state government jobs in Punjab on women's labour market outcomes. Specifically, it examines the effect on the share of women in government jobs.

2 Data

The paper uses data from the Periodic Labour Force Survey (PLFS), a state-conducted survey that captures individual-level information on employment, labour market participation, and demographics. The data is available quarterly from 2017 to 2024.

Due to the seasonal nature of jobs, especially in agriculture, the data is aggregated to the annual level in order to smooth short-run fluctuations. Further, 2017 and 2024 are each only observed for two quarters, therefore, these years are excluded. The sample is restricted to individuals aged 21–47, which corresponds to the eligibility age range for direct recruitment across all four service groups. Sampling weights from PLFS are applied throughout.

Individuals with a government job are identified using the enterprise type of principal activity[3] as a public sector establishment. The main outcome variable of interest is the share of women in government jobs.

[1] "Public sector establishments" refer to any industry, trade, business, or occupation owned, controlled, or managed by the State Government, as well as government companies in which at least 51% of the paid-up share capital is held by the Government of Punjab, as defined under Section 2(45) of the Companies Act, 2013.

[2] Government posts in India are classified into four groups based on seniority and responsibilities. Group A includes senior administrative and managerial posts, Group B includes middle-level administrative positions, Group C consists largely of clerical and technical staff, and Group D includes support and operational roles.

[3] The activity on which a person spent a relatively longer time during the reference period of the last 365 days preceding the survey.

3 Empirical Strategy

To estimate the causal impact of the 33% reservation for women in public sector jobs in Punjab, this paper uses the synthetic control method (SCM). States with similar policies introduced earlier or during this period are removed from the donor pool.[4]

Let Y_{it} denote the outcome of interest for state i in year t. The reservation policy was announced in October 2020, therefore 2021 is considered the first treatment year. Let $T_0 = 2020$ denote the last pre-treatment period. The treatment indicator is defined as

$$D_{it} = \begin{cases} 1 & \text{if } i = Punjab \text{ and } t \geq 2021 \\ 0 & \text{otherwise.} \end{cases}$$

The estimated treatment effect is calculated as the difference between the observed outcome in Punjab and the weighted average of outcomes in the donor states:

$$\hat{\tau}_t = Y_{Punjab,t} - \sum_{j=1}^{J} w_j Y_{j,t}, \quad t > T_0.$$

4 Results

Figure 3 compares the observed share of women in government jobs in Punjab with its synthetic control. During the pre-treatment period, the two trajectories closely track each other, indicating that the synthetic control provides a reasonable approximation of the counterfactual.

Following the policy's introduction in 2021, the trajectories begin to diverge. Punjab experiences a small short-lived decline. By 2023, the share of women in government jobs in Punjab reaches approximately 0.46, while the synthetic control remains around 0.36–0.37.

Placebo tests are conducted by applying the synthetic control method to each donor state. As shown in Figure 5, Punjab's post-treatment gap is larger than most placebo gaps. Following Abadie et al. (2010), states with poor pre-treatment fit are excluded by discarding placebo units whose pre-treatment RMSPE exceeds twenty times that of Punjab. The pattern remains similar after applying this restriction (Figure 6), suggesting that the observed increase is unlikely to be driven by random variation.

5 Conclusion

This paper evaluates the impact of Punjab's 33% reservation policy for women in government jobs using the synthetic control method. The results suggest that the policy is associated with a substantial increase in the share of women employed in government jobs. Placebo tests indicate that the estimated effect is large.

However, the results should be interpreted with caution due to the short pre-treatment period available in the PLFS data. Future work could examine longer time horizons and additional labour market outcomes, such as female labour force participation, sectoral employment patterns, education rates as well as spillovers in private sector.

[4] Bihar, Chattisgarh, Madhya Pradesh, Andhra Pradesh and Telangana

A Appendix: Tables and Figures

Table 1: Share of women in government jobs predictor means

	Punjab	Synthetic Punjab	Average of control states
Women Share 2018	0.32	0.32	0.31
Women Share 2019	0.35	0.35	0.34
Women Share 2020	0.37	0.37	0.34
Age	33.22	33.26	33.33
Low Education	0.17	0.15	0.14
Medium Education	0.45	0.53	0.53
High Education	0.30	0.27	0.28
Rural	0.62	0.59	0.61
LFPR	0.24	0.35	0.38

Table 2: Synthetic control donor weights

State	State Code	Weight
Jammu & Kashmir	01	0.027
Himachal Pradesh	02	0.036
Chandigarh	04	0.028
Uttrakhand	05	0.030
Haryana	06	0.028
Delhi	07	0.028
Rajasthan	08	0.031
Uttar Pradesh	09	0.028
Sikkim	11	0.036
Arunachal Pradesh	12	0.026
Nagaland	13	0.032
Manipur	14	0.027
Mizoram	15	0.030
Tripura	16	0.038
Meghalaya	17	0.044
Assam	18	0.031
West Bengal	19	0.032
Jharkhand	20	0.031
Odisha	21	0.040
Gujarat	24	0.031
Daman & Diu	25	0.058
Maharashtra	27	0.029
Karnataka	29	0.034
Goa	30	0.033
Lakshadweep	31	0.029
Kerala	32	0.076
Tamil Nadu	33	0.044
Puducherry	34	0.029
A & N Island	35	0.033

Figure 1: Gender composition of government jobs in Punjab

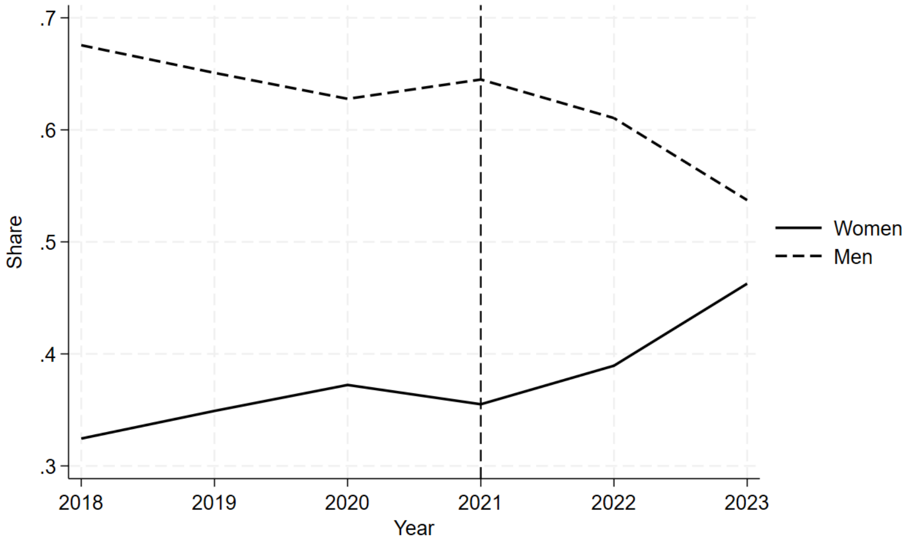


Figure 2: Trends in share of women in government jobs in Punjab and average of control states

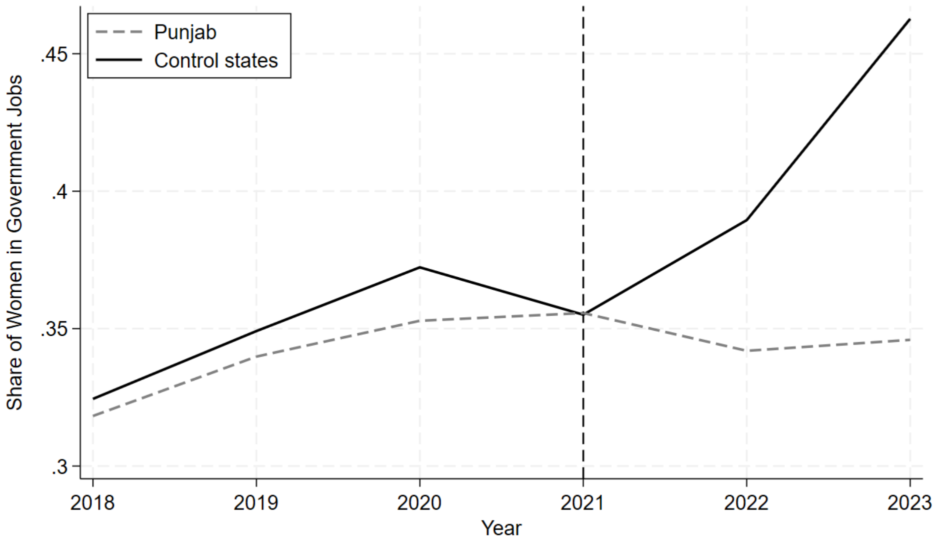


Figure 3: Women's Share in Government Jobs in Punjab and synthetic Punjab

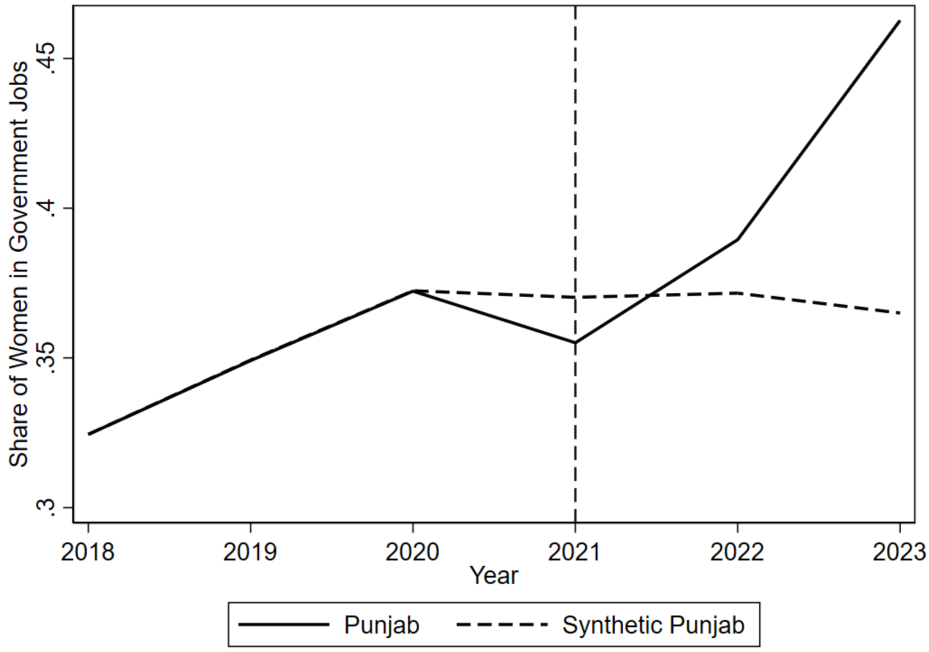


Figure 4: Gap in share of women in government jobs between Punjab and synthetic Punjab.

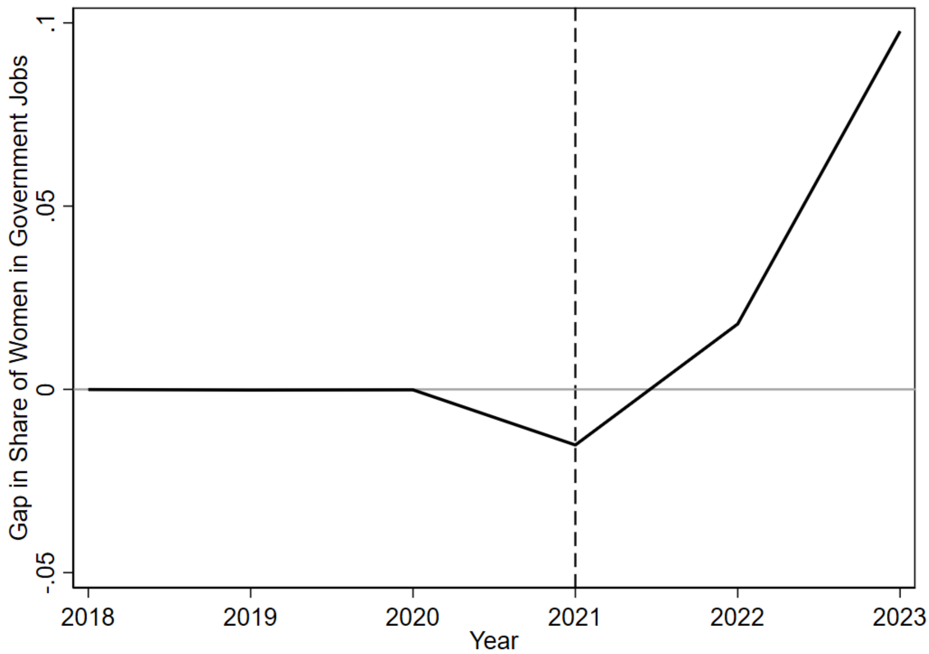


Figure 5: Share of women in government jobs in Punjab and placebo gaps in all 29 control states.

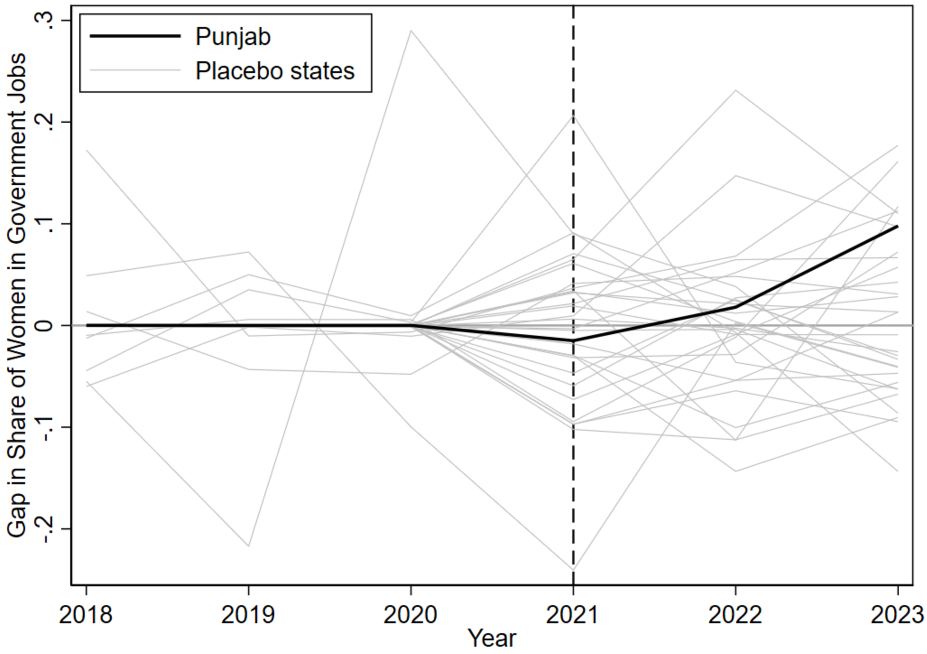
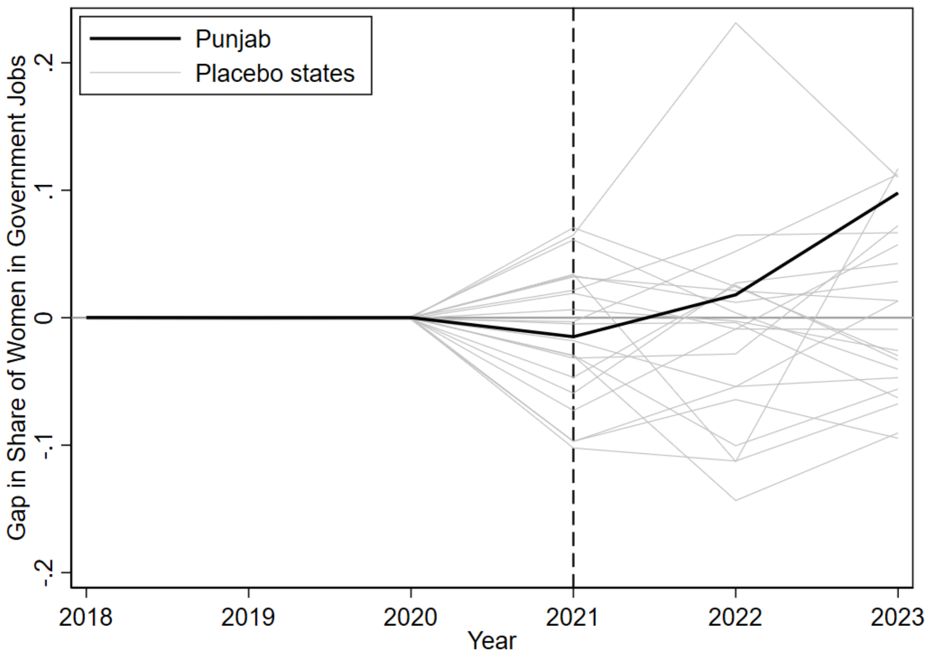


Figure 6: Share of women in government jobs in Punjab and placebo gaps in all 22 control states (discards states with pre-Proposition MSPE 20 times higher than Punjab's) .



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